

PROCUREMENT DATA UPDATE AUTOMATION

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Work 1:

Automated PR information collection using Power Automate

Objective

Automatically collects Purchase Requisition (PR) information on Outlook email and maps onto an Excel file by deploying **Power Automate** and dynamic expressions.

Implementation timeline

In total 2 weeks:

- Week 1: Business assessment, pain point identification, and prototype development.
- Week 2: Pilot testing and user feedback collection; then conduct prototype refinement.

Tools used

Microsoft 365 apps: Power Automate, Outlook, Excel

Impacts

- Eliminated 100% manual effort required to collect and record PR information from email notifications, *though it does not capture the PR's detailed line item specifications*
- Reduced data entry errors caused by manual copying, since the flow extracts data directly from the email rather than retyping it.
- Saved approximately 1 to 2 working hours per week for Buyers, allowing more time for sourcing and supplier management activities.

Problem statement

A Buyer is _____ overwhelmed by at least 200 PRs per month

who needs _____ a fast, reliable way to gather PR information into one Excel file

because _____ manually copying PR details from each incoming email into Excel is repetitive and error prone,

_____ especially at this volume.

Solution Design



Overview: The tool listens for incoming PR notification emails and automatically extracts the relevant fields into a tracking file, removing the need for buyers to open and read each email manually.

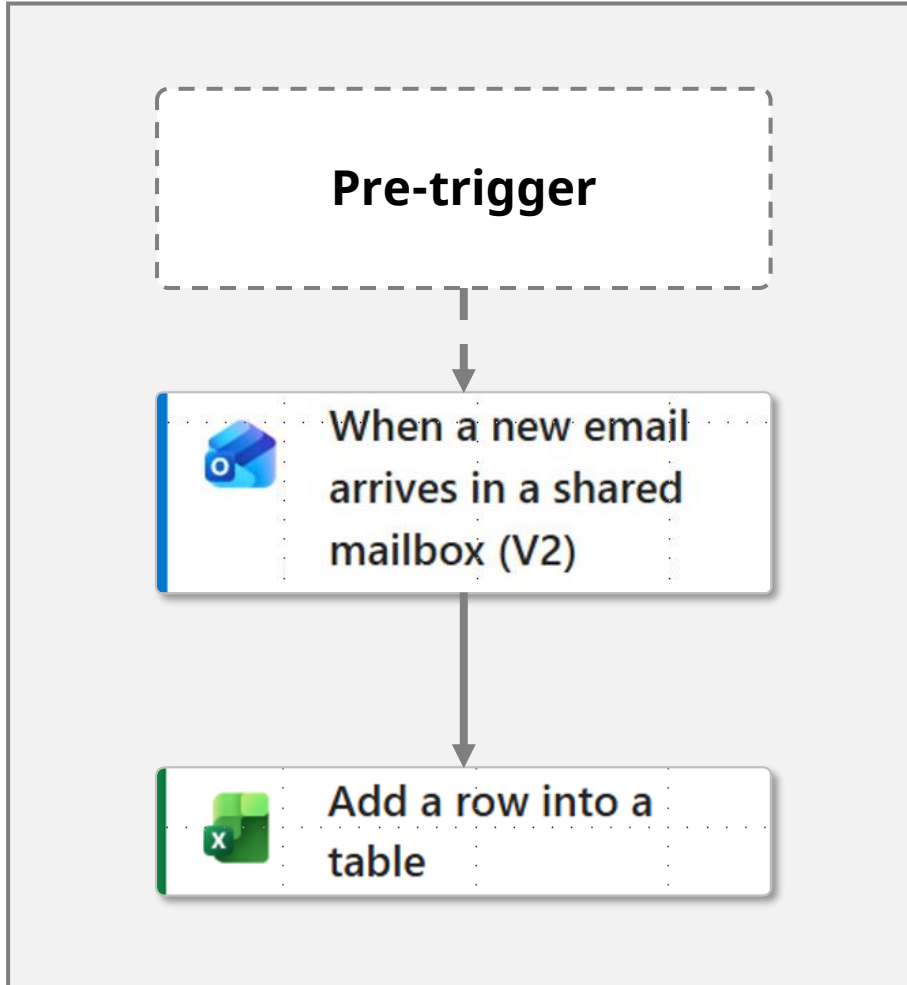


Key challenge: PR information only exists as unstructured text inside the email subject and sender address, not as a structured field.



Idea: Use dynamic expressions to parse fixed patterns in the email (prefixes, tags, delimiters) into structured columns.

Solution Design



Flowchart: Workflow in Power Automate

0 Pre-trigger:

User fills out a form for required information of a PR on BASE platform.

1 Trigger:

When a new email arrives in a shared mailbox (V2)

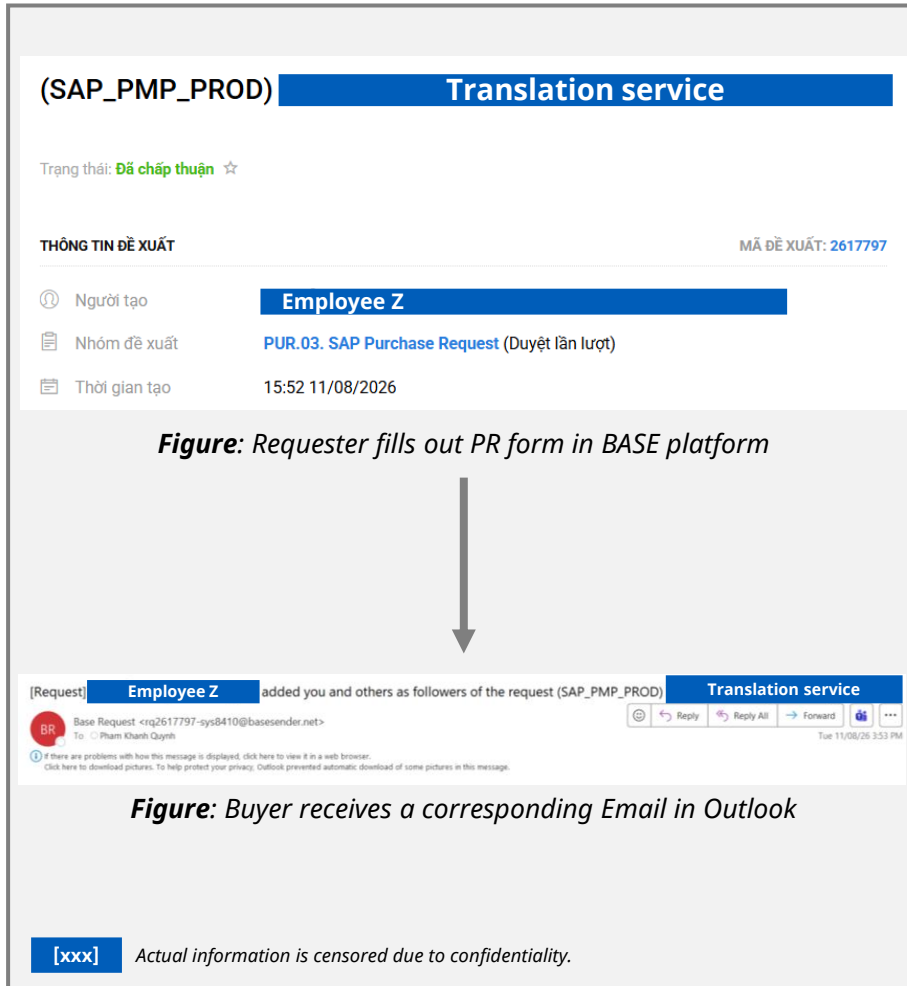
- Application: Outlook 365
- Mailbox: [Email of buyer]

2 Action:

Add a row into a table

- Application: Excel 365
- Extract the information from the email and fills out 'BASE Request.xlsx'

Solution Design



Flowchart: PR form in BASE platform and corresponding Email in Outlook

0

Pre-trigger:

User fills out a form for required information of a PR on BASE platform.

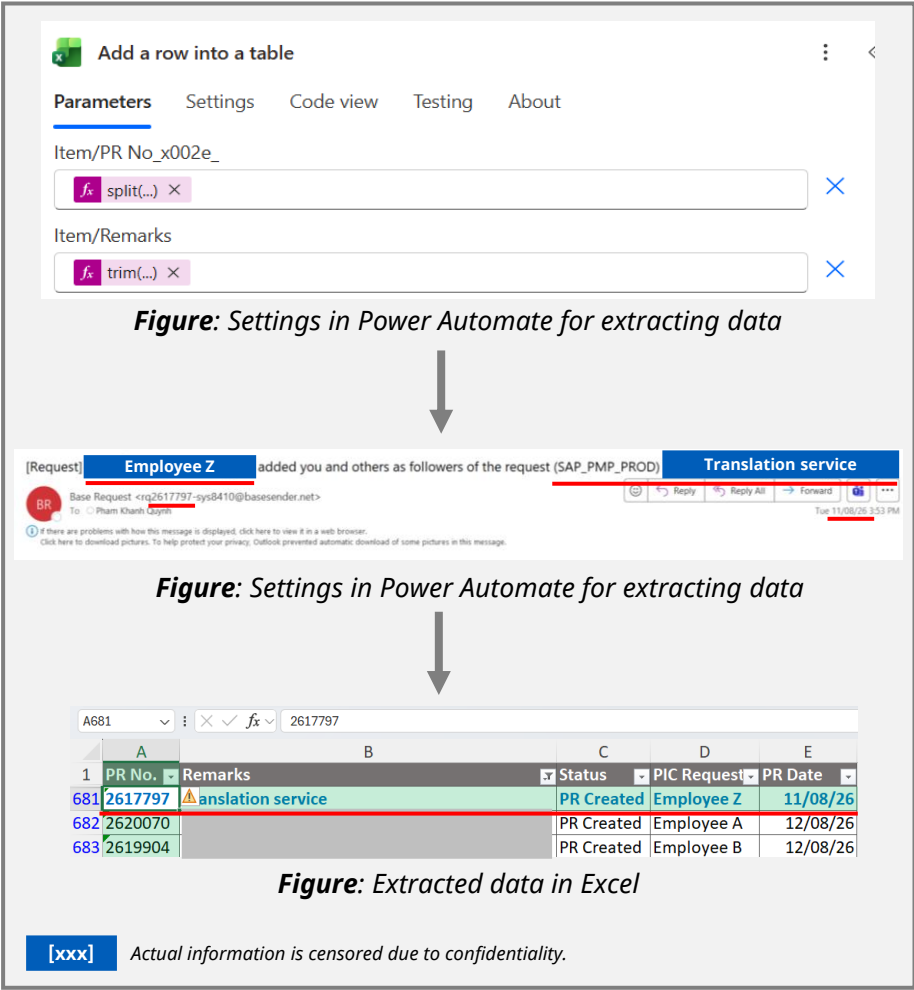
1

Trigger:

When a new email arrives in a shared mailbox (V2)

- Application: Outlook 365
- Mailbox: [Email of buyer]

Solution Design



Flowchart: Extracted data from Email to Excel

2 Action: Add a row into a table

- Application: Excel 365
- Extract the information from the email and fills out 'BASE Request.xlsx'

Field	Source	Dynamic Expressions	Output Example	Technical Explanation
PR ID	Sender Email Address	split(split(triggerOutputs()?['body/ from'],'rq')[1],'-sys8410@basesender.net')[0]	2617797	Extract the characters between the predefined email prefix 'rq' and suffix '-sys8410@basesender.net'
PR Name	Email Subject	trim(substring(triggerOutputs()?['body/subject'], add(indexOf(triggerOutputs()?['body/subject'], '(SAP_PMP_PROD)'), length('(SAP_PMP_PROD')))))	Translation service	Extract the characters after the system tag '(SAP_PMP_PROD)' within the email subject and captures all text that follows it.
Status	Default Value	PR Created	PR Created	A predefined status is automatically assigned whenever a new PR record is created in the tracking database.
PIC Requester	Email Subject	trim(substring(triggerOutputs()?['body/subject'], add(indexOf(triggerOutputs()?['body/subject'], '[Request] '), length('[Request] ')), sub(indexOf(triggerOutputs()?['body/subject'], 'added you and others as followers of the request'), add(indexOf(triggerOutputs()?['body/subject'], '[Request] '), length('[Request] '))))))	Employee Z	The flow identifies the text located between the request notification prefix '[Request]' and the system-generated message within the email subject 'added you and others as followers of the request'.
PR Date	Email Received Timestamp	formatDateTime(triggerOutputs()?['body/receivedDateTime'], 'yyyy-MM-dd')	2026-11-08	The flow retrieves the email received timestamp from Outlook and converts it into a standardized date format 'yyyy-MM-dd') before storing it in the tracking file.

Semi-automated PR-PO reconciliation using Power Query

Objective

Automatically consolidates and matches Purchase Request (PR) and Purchase Order (PO) data from SAP and BASE into a master tracking file, by deploying **Power Query and table join techniques**, while preserving existing tracking notes unaffected by each data refresh.

Implementation timeline

In total 4 weeks:

- Week 1: Business assessment, pain point identification, and prototype development.
- Week 2: Pilot testing and user feedback collection.
- Week 3: Prototype refinement based on pilot results.
- Week 4: Solution deployment, performance monitoring, and continuous improvement planning.

Tools used

Microsoft 365 apps: Excel, Power Query (M language), table joins (full outer join, left anti join), data cleaning and transformation.

Impacts

- Eliminated 99% manual cross checking between two separate systems (SAP and BASE) to determine PR to PO status, *though matching accuracy still depends on the PR number being entered correctly in SAP at PO creation.*
- Preserved buyers' historical tracking notes that were previously at risk of being lost on each data refresh.

Problem statement

A Buyer is _____ overwhelmed by high volume PR and PO across two disconnected systems

who needs _____ a fast, reliable way to see which PR already has a matching PO and which are still pending

because _____ manually cross-checking SAP and BASE, then keeping tracking notes updated by hand,

_____ is time consuming and easy to get wrong as volume grows.

Solution Design



Overview: The tool combines PR data from BASE and PO data from SAP into a single reconciled view using Power Query, refreshed with one click instead of manual lookup. It is built to keep working the same way as data volume grows, and to never lose a buyer's tracking notes even as the underlying data refreshes.

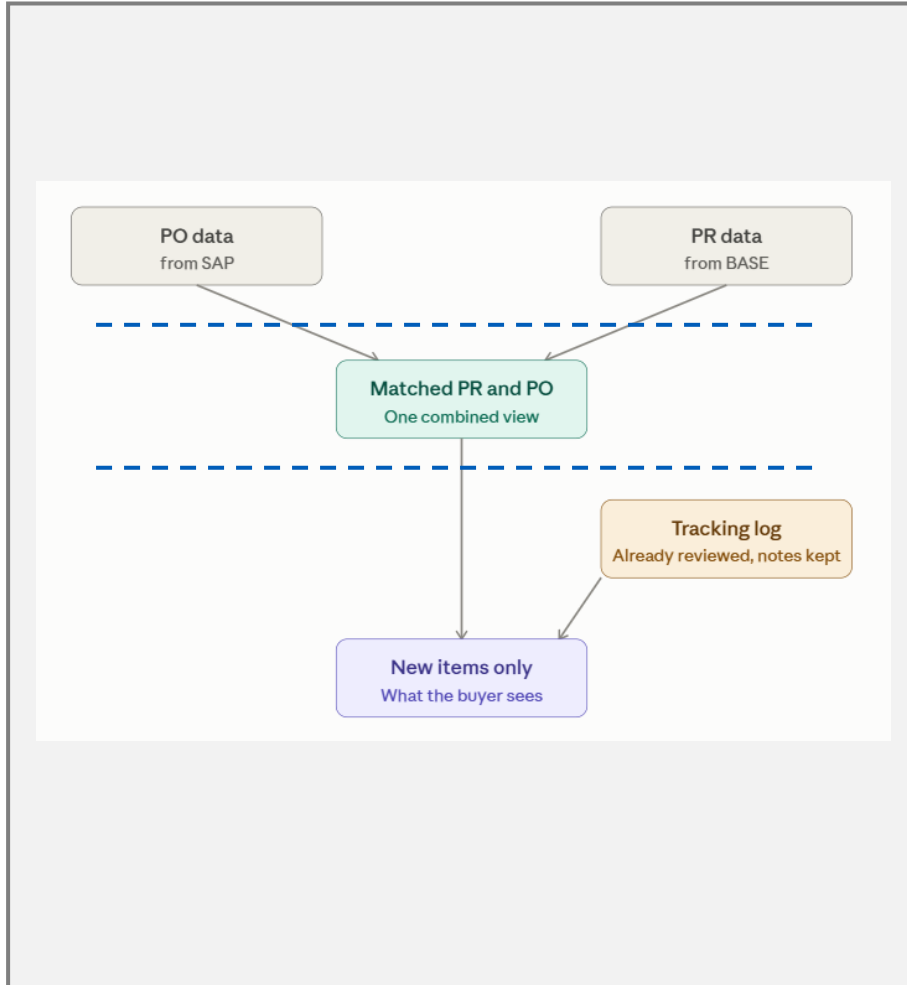


Key challenge: Every refresh would wipe out any notes the buyer had already added, since the source data reloads completely each time. The design challenge was automating the repetitive matching work without losing the manual notes buyers had already added.



Idea: Splitting the workflow into a live matching layer and a separate persistent tracking layer solved this: buyers get the speed of automation without losing the context they build up over time.

Solution Design



Flowchart: PR to PO mapping flow

1

Source layer: 2 queries

Pull raw data directly from two disconnected systems: PO records from a SAP export and PR records from a BASE export.

2

Processing layer: 1 query

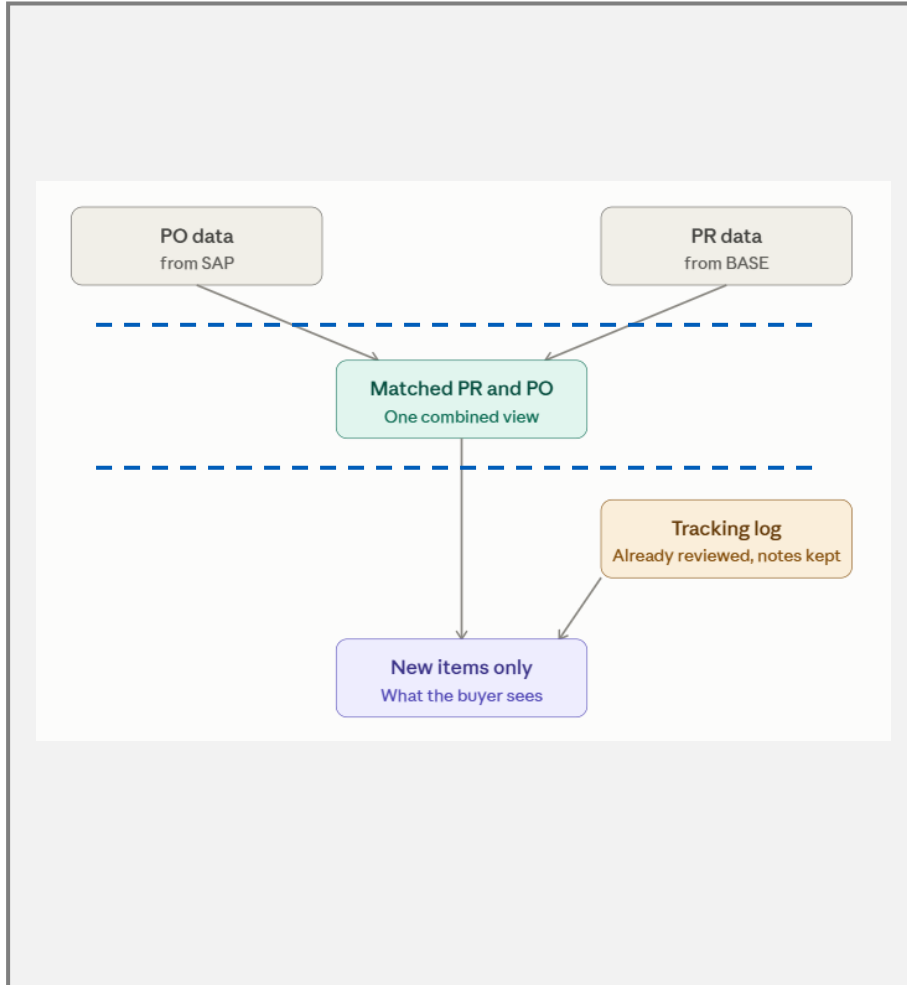
Join both datasets on PR number into a single combined table, showing every PR alongside its matching PO if one already exists.

3

Comparison layer: 2 queries

Compare the combined table against a running tracking log, then surface only the records that are new since the last check. Previously reviewed records, along with their notes, stay untouched.

Solution Design



Flowchart: PR to PO mapping flow

1

Source layer — 2 queries

Pull raw data directly from two disconnected systems: PO records from a SAP export and PR records from a BASE export.

'PR Data': is BASE Request.xlsx from Work 1.

'PO Data':

- Extract from txt.file which is exported directly from SAP.
- Extract the PR number from the Remarks. Since SAP has no dedicated PR column; the PR only exists as free text typed in Remarks by whoever created the PO.

Solution Design

	ABC 123 PR_PO key	ABC 123 PR	ABC 123 PO	ABC 123 Remarks
1	2570837_140008141	2570837	140008141	
2	2575079_140007959	2575079	140007959	
3	2569534_140007943	2569534	140007943	
4	2564035_140008140	2564035	140008140	
5	2596753	2596753		null
6	2595358	2595358		null
7	2595374	2595374		null
8	2502335_140007946	2502335	140007946	
9	2551500_140008133	2551500	140008133	
10	2564124_140007925	2564124	140007925	
11	2564103_140007923	2564103	140007923	
12	2379928_140007956	2379928	140007956	
13	2379910_140007955	2379910	140007955	
14	2501669_140008095	2501669	140008095	
15	2586070	2586070		null
16	2586070	2586070		null
17	2363107_140008132	2363107	140008132	
18	2419405_140007949	2419405	140007949	
19	2551517_140007958	2551517	140007958	
20	2539069_140007957	2539069	140007957	
21	10.2026_140007920	10.2026	140007920	
22	_9.2026_140007919	_9.2026	140007919	
23	_7.2026_140007922	_7.2026	140007922	
24	2521069_140007906	2521069	140007906	
25	2563023_140007889	2563023	140007889	
26	2580059_140008142	2580059	140008142	
27	2580029_140008001	2580029	140008001	

Figure: 'Matched PR-PO' query table with every PR alongside its matching PO if one already exists.

2

Processing layer — 1 query

Join both datasets on PR number into a single combined table, *showing every PR alongside its matching PO if one already exists.*

Key transformations:

- Full outer join between the 2 tables from Sourcing layer on the PR column.
- Create a composite key column 'PR_PO key' (PR combined with PO) to uniquely identify each PR to PO pairing.

Note: 1 PR can generate more than 1 PO, but 1 PO is originated from only 1 PR.

Solution Design

PR PO key	PR	PO	Remarks	BP Name	PO Value	PR Date	PO Date	Requester	User	Status	Last updated
2574373_140008240	2574373	140008240				7/16/2026	8/6/2026			PO Approv	8/6/2026
2585858_140008200	2585858	140008200				7/23/2026	8/5/2026			PO Approv	8/5/2026
2587088_140008199	2587088	140008199				7/24/2026	8/5/2026			PO Approv	8/5/2026
2597678_140008241	2597678	140008241				7/30/2026	8/4/2026			PO Approv	8/4/2026
2578636_140008243	2578636	140008243				7/31/2026	8/6/2026			PO Approv	8/6/2026
2582044_140008208	2582044	140008208				7/31/2026	8/5/2026			PO Approv	8/5/2026
2582044_140008212	2582044	140008212				7/31/2026	8/5/2026			PO Approv	8/5/2026
2582044_140008213	2582044	140008213				7/31/2026	8/5/2026			PO Approv	8/5/2026
2582044_140008349	2582044	140008349				7/31/2026	8/12/2026			PO Approv	8/12/2026
2604556_140008331	2604556	140008331				8/3/2026	8/11/2026			PO Approv	8/11/2026
2603387_140008349	2603387	140008349				8/10/2026	8/12/2026			PO Approv	8/12/2026
2617797_140008362	2617797	140008362				8/11/2026	8/12/2026			PO Approv	8/12/2026
9_2026_140008174	9_2026	140008174				8/3/2026				Generated	8/3/2026
9_2026_140008173	9_2026	140008173				8/3/2026				Generated	8/3/2026
9_2026_140008172	9_2026	140008172				8/3/2026				Generated	8/3/2026
9_2026_140008171	9_2026	140008171				8/3/2026				Generated	8/3/2026
9_2026_140008271	9_2026	140008271				8/7/2026				PO Approv	8/7/2026
9_2026_140008272	9_2026	140008272				8/7/2026				PO Approv	8/7/2026
9_2026_140008273	9_2026	140008273				8/7/2026				PO Approv	8/7/2026
9_2026_140008278	9_2026	140008278				8/7/2026				PO Approv	8/7/2026
9_2026_140008300	9_2026	140008300				8/10/2026				PO Approv	8/10/2026
10_2026_140008202	10_2026	140008202				8/5/2026				PO Approv	8/5/2026
10_2026_140008279	10_2026	140008279				8/7/2026				PO Approv	8/7/2026
11_2026_140008345	11_2026	140008345				8/12/2026				PO Approv	8/12/2026
12_2026_140008346	12_2026	140008346				8/12/2026				Generated	8/12/2026

Figure: 'New items only' query table

Items from 'New items only' are then copied to 'Tracking logs'

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	PR_PO key	PR	PO	Remarks	BP Name	PO Value	PR Date	PO Date	Requester	User	Status	Last updated	Notes	Thống kê hàng dự	Ngày giao hàng dự	PO sent to customer	PO sent to company internal account	PO sent to company internal account	PO sent to Accounting	Accounting PO	Complete Decs. sent to customer	BS check
2	2582044_140008349	2582044	140008349				31/07/26	12/08/26			PO Approved	12/08/26										
4	2603387_140008349	2603387	140008349				10/08/26	12/08/26			PO Approved	12/08/26										
6	2617797_140008362	2617797	140008362				11/08/26	12/08/26			PO Approved	12/08/26										
8	2619804	2619804					12/08/26				PR Open	12/08/26										
7	2620070	2620070					12/08/26				PR Open	12/08/26										
9	2604556_140008331	2604556	140008331				03/08/26	11/08/26			PO Approved	11/08/26										
10	2610000	2610000					11/08/26				PR Open	11/08/26										
11	2610029	2610029					11/08/26				PR Open	11/08/26										
12	2617108	2617108					11/08/26				PR Open	11/08/26										
13	2603387	2603387					10/08/26				PR Open	10/08/26										
14	2608300	2608300					10/08/26				PR Open	10/08/26										

Figure: 'Tracking logs' table

Flowchart:

3

Comparison layer — 2 queries

Compare the combined table against a running tracking log, then surface only the records that are new since the last check. Previously reviewed records, along with their notes, stay untouched.

'Tracking logs': Is updated manually by copying rows from 'New items only' and pasting them here after each review.

'New items only':

- Filters out exactly the PR to PO records that are new and not yet noted.
- Key transformations:
 - Left anti join 'Matched PR-PO' against 'Tracking logs' on 'PR_PO key' to keep only the rows present in 'Matched PR-PO' but not yet in 'Tracking logs'.
 - Add a 'Last updated' column (PO Date takes priority; if no PO yet, falls back to PR Date)



Both tools depend on free text entered by humans (email subject/sender in Work 1, SAP Remarks in Work 2) rather than a structured field. This is the main risk in both: a changed template or a mistyped PR number can silently break the extraction or matching.

Automation is only as complete as the data it can access. Work 1 captures PR ID, requester, and date, but not the PR's detailed specification, since that was never in the email.

Separating live refreshed data from a manually maintained tracking layer (Work 2) let automation and manual notes coexist, a pattern reusable for future tools with recurring data and human input to preserve.

Lessons learned

1

Both tools depend on free text entered by humans (email subject/sender in Work 1, SAP Remarks in Work 2) rather than a structured field. This is the main risk in both: a changed template or a mistyped PR number can silently break the extraction or matching.

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Separating live refreshed data from a manually maintained tracking layer (Work 2) let automation and manual notes coexist, a pattern reusable for future tools with recurring data and human input to preserve.

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THANK YOU